CompTIA Security+ Exam SY0-701

Lesson 1

Summarizing Fundamental Security Concepts

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Objectives

- Summarize information security concepts
- Compare and contrast security control types
- Describe security roles and responsibilities

Lesson 1

Topic 1A

Security Concepts

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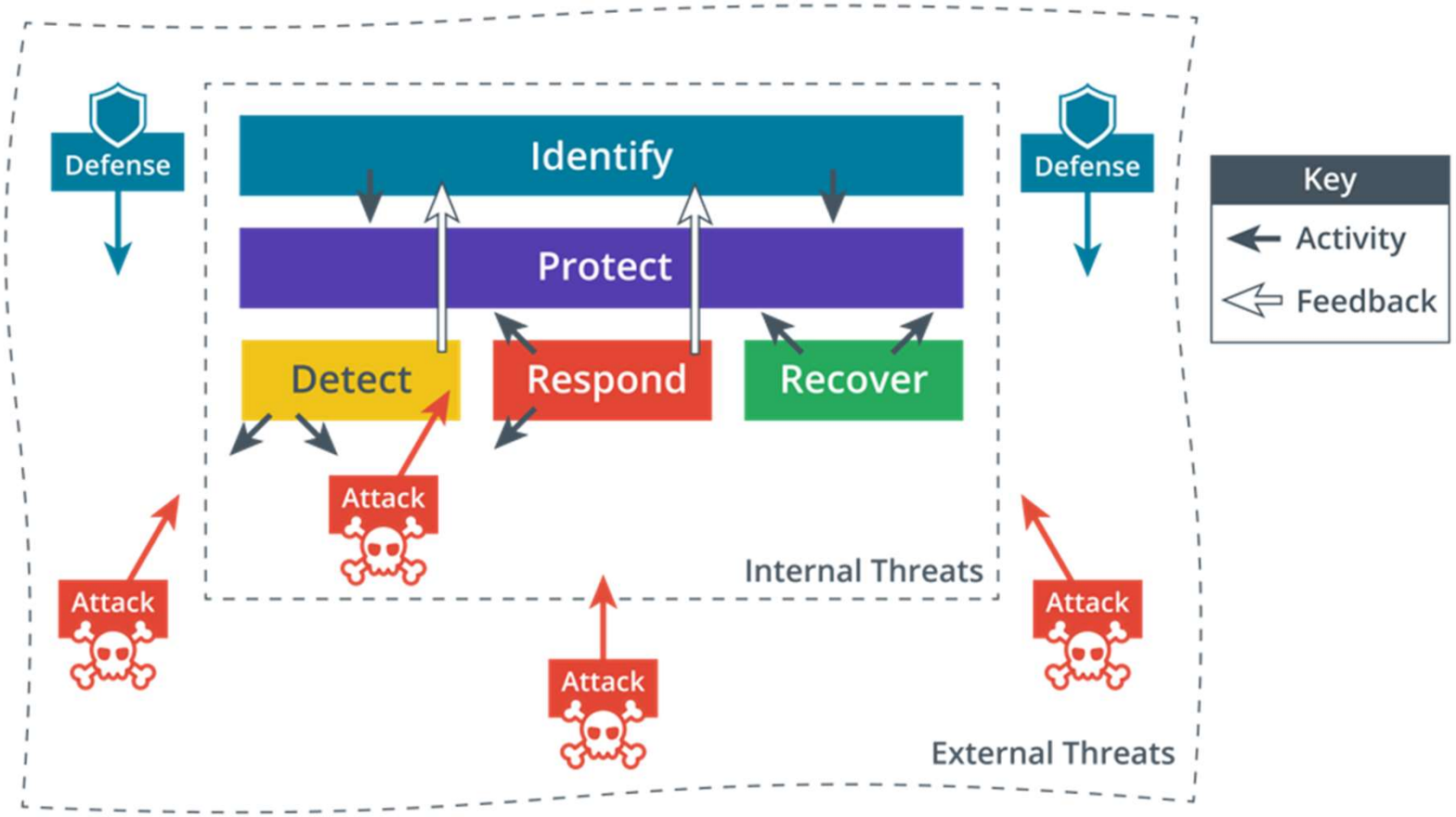
Information Security

- Confidentiality
  - Information should only be read by authorized persons
- Integrity
  - Data is stored and transferred as intended and any modification is authorized
- Availability
  - Information is accessible to those authorized to view or modify it
- Non-repudiation
  - Persons cannot deny creating or modifying data

Cybersecurity Framework



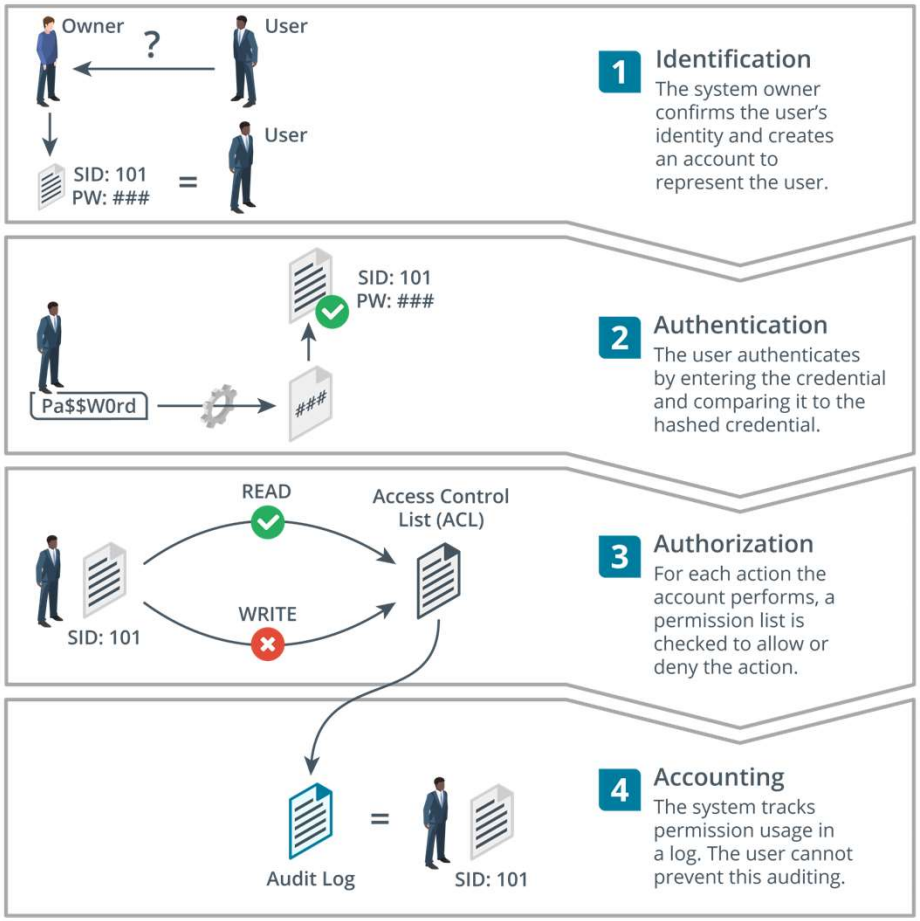
Gap Analysis

current state
vs desired
state

Function	Controls (Actual/Required)	CIA Triad Risk Levels	Target Remediation
Identify (10/16)	Asset Management (4/6)	C : 6 I : 6 A : 6	Q4
	Governance (3/4)	C : 6 I : 6 A : 1	Q3
	Risk Assessment (3/6)	C : 6 I : 6 A : 3	Q3
Protect (8/16)	Identity and Access Management (5/8)	C : 9 I : 9 A : 4	Q1
	Data Security (3/8)	C : 9 I : 9 A : 4	Q1

- Advanced capability
- Intermediate capability
- No/basic capability

Access Control



Review Activity: Security Concepts

- Information security
  - CIA triad
- Cybersecurity framework
- Gap analysis
- Access control
  - IAM and AAA

Lab Activity

- Assisted Lab: Exploring the Lab Environment
- Assisted Lab: Perform System Configuration Gap Analysis

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Topic 1B

Security Controls

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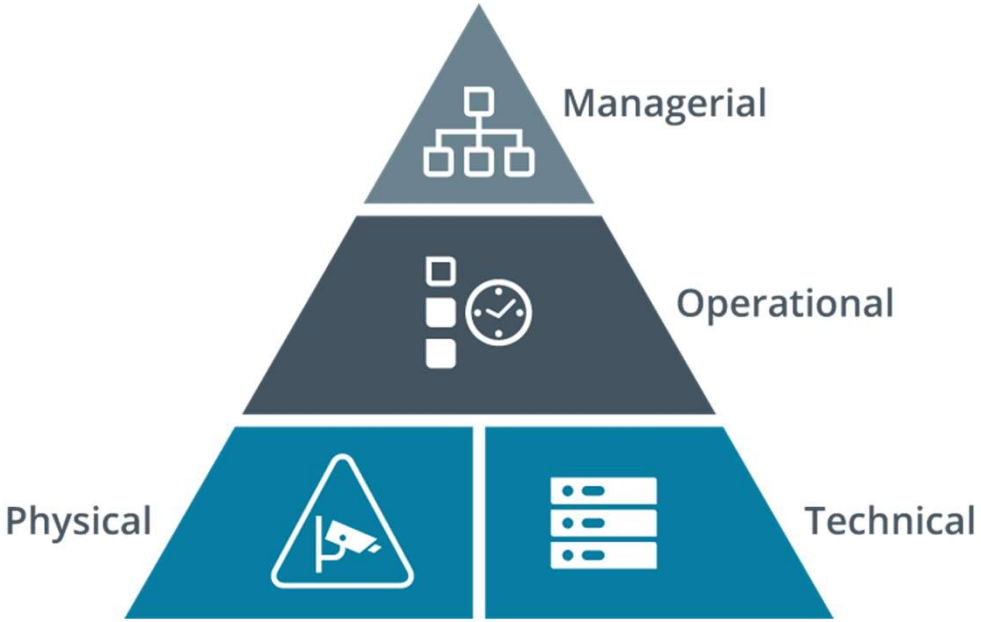
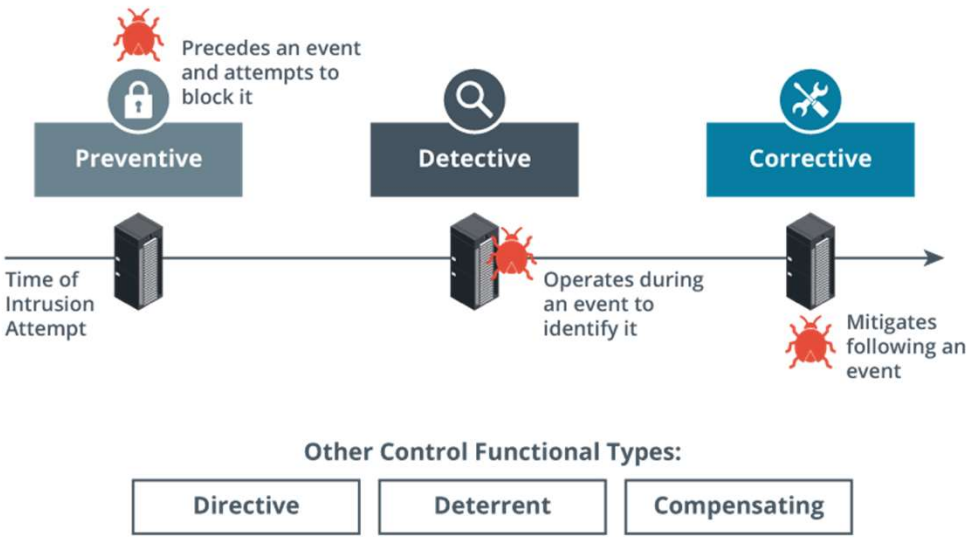
Security Control Categories

- Managerial
  - Give oversight of system
- Operational
  - Relies on a person for implementation
- Technical
  - Implemented in operating systems, software, and security appliances
- Physical
  - Devices that mediate access to premises and hardware



Security Control Functional Types (1)



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- Preventive
  - Physically or logically restricts unauthorized access
  - Operates before an attack
- Detective
  - Identifies attempted or successful intrusions
  - Operates during an attack
- Corrective
  - Responds to and fixes an incident and may prevent its reoccurrence
  - Operates after an attack

Security Control Functional Types (2)

- Directive
  - Enforces a rule of behavior
- Deterrent
  - Psychologically discourages intrusions
- Compensating
  - Substitutes for a principal control
  - Associated with framework compliance measures

Information Security Roles and Responsibilities

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- Overall responsibility
  - Chief Information Officer (CIO)
  - Chief Security Officer (CSO)
- Managerial
- Technical
  - Information Systems Security Officer (ISSO)
- Non-technical
- Due care/liability

Information Security Competencies

- Risk assessments and testing
- Specifying, sourcing, installing, and configuring secure devices and software
- Access control and user privileges
- Auditing logs and events
- Incident response and reporting
- Business continuity and disaster recovery
- Security training and education programs

Information Security Business Units

- Security Operations Center (SOC)
- DevSecOps
  - Development, security, and operations
- Incident response
  - Cyber incident response team (CIRT)

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Review Activity: Security Controls

- Security control categories
  - Managerial, operational, technical, physical
- Security control functional types
  - Preventive, detective, corrective plus directive, deterrent, compensating
- Information security roles and responsibilities
- Information security competencies
- Information security business units
  - SOC, DevSecOps, and CIRT

Lab Activity

- Assisted Lab: Configuring Examples of Security Control Types

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Summary

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